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Editor-in-Chief

Journal of Construction Engineering and Management

Dear Editor,

We are pleased to submit our manuscript entitled “**A Hybrid Intelligent Framework for Building Retrofitting Cost Analytics: Integrating Segmented Classification with Transfer Learning-based Regressors**” for consideration for publication in the *Journal of Construction Engineering and Management*. This manuscript is authored by Nahyun Kwon, Sojin Park, Yonghan Ahn, and Ho Anh Thu Nguyen.

Accurate early-stage cost estimation is a fundamental challenge in construction engineering and management, particularly for building retrofitting projects where historical data are limited and project costs are highly heterogeneous. Existing machine learning-based cost prediction models often struggle to generalize to high-cost projects because such cases are underrepresented in available datasets. Although some previous studies mitigate this issue by excluding extreme-cost projects during preprocessing, this practice introduces sampling bias and limits the practical applicability of predictive models for construction decision-making.

To address these challenges, we propose a hybrid intelligent framework that integrates Support Vector Machine (SVM)-based cost-range classification with segment-specific Artificial Neural Network (ANN) regression models enhanced through transfer learning. By transferring knowledge from data-rich low-cost segments to data-scarce high-cost segments, the proposed framework effectively addresses data imbalance while preserving valuable information from extreme-cost projects. This design improves predictive robustness across the entire project cost spectrum and provides a practical solution for early-stage construction cost estimation.

The proposed framework was evaluated using a real-world dataset comprising 419 building retrofitting projects. Through systematic investigation of multiple cost segmentation strategies, we identified an optimal classification threshold that maximized predictive performance. Compared with conventional single-stage machine learning models, the proposed approach achieved an overall R^2 of 0.96 while reducing the MAPE and RMSE by 46% and 25%, respectively. Furthermore, SHAP-based model interpretation was employed to identify the primary cost-driving factors, providing transparent and explainable insights to support construction planning and cost management.

We believe this work aligns closely with the aims and scope of the *Journal of Construction Engineering and Management*. Specifically, the manuscript contributes to several core areas emphasized by the journal, including construction estimating, cost control, construction management, and computer applications in construction engineering. Beyond its methodological contribution, the proposed framework offers a practical decision-support tool that improves the reliability of early-stage cost estimation for retrofit projects under limited and imbalanced data conditions, making it directly relevant to both researchers and practitioners in construction engineering and management.

In addition, this research supports the United Nations Sustainable Development Goals, particularly Industry,

Innovation and Infrastructure through the development of intelligent digital technologies for construction management, and Sustainable Cities and Communities by facilitating more efficient planning and management of building retrofit projects that contribute to the sustainability and resilience of the built environment.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the submitted manuscript, and the authors declare that there are no conflicts of interest.

Thank you for your time and consideration. We appreciate your evaluation of our manuscript and hope it will be considered suitable for publication in the *Journal of Construction Engineering and Management*.

Sincerely,

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